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ChurnZero 26

Customer Churn Prediction System for Banking Retention

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Problem Statement:

Banks lose 15-25% of customers annually to churn, directly impacting deposits, cross-sell revenue, and lifetime value.

So we build a predictive model on a banking dataset to identify at-risk customers and propose data-driven retention strategies that a business team could actually execute.

What is Churn?

Customer churn is basically a breakup between the customer and the bank.

Why Does It Hurt Banks?

The bank loses future revenue, trust, and acquisition cost together.

Why Prediction Matters?

Customer churn prediction transforms banks from reactive damage control to proactive customer retention.

OBJECTIVES

- Predict customer churn accurately
- Identify key churn drivers
- Improve customer retention strategies
- Provide explainable insights

DATASET OVERVIEW

▶

df.columns

↑ ↓ 🗑

[32...

Index(['CreditScore', 'Geography', 'Gender', 'Age', 'Tenure', 'Balance',
 'NumOfProducts', 'HasCrCard', 'IsActiveMember', 'EstimatedSalary',
 'Exited'],
 dtype='object')

+ Code

+ Markdown

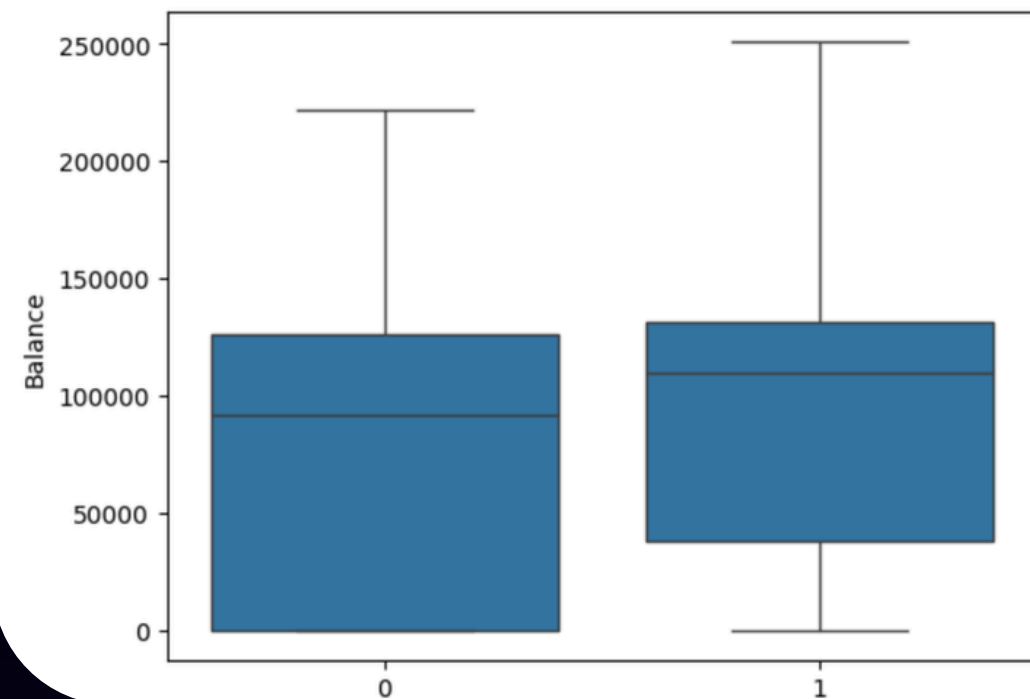
Column	Meaning
CreditScore	Customer’s creditworthiness score
Geography	Country/location of customer
Gender	Customer gender
Age	Customer age
Tenure	Number of years customer stayed with bank
Balance	Current bank account balance
NumOfProducts	Number of bank products used by customer
HasCrCard	Whether customer owns a credit card
IsActiveMember	Whether customer actively uses bank services
EstimatedSalary	Estimated annual salary of customer
Exited	Target column – whether customer churned/left the bank (1=yes, 0=no)

KEY INSIGHTS

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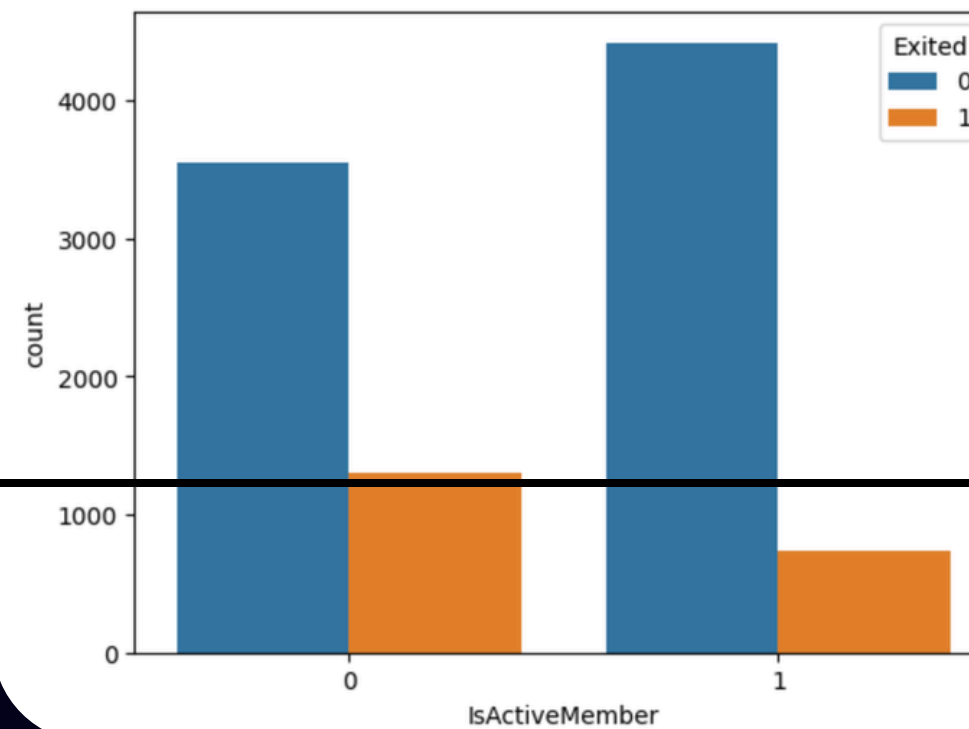
```
sns.boxplot(x='Exited', y='Balance', data=df)
```

<Axes: xlabel='Exited', ylabel='Balance'>



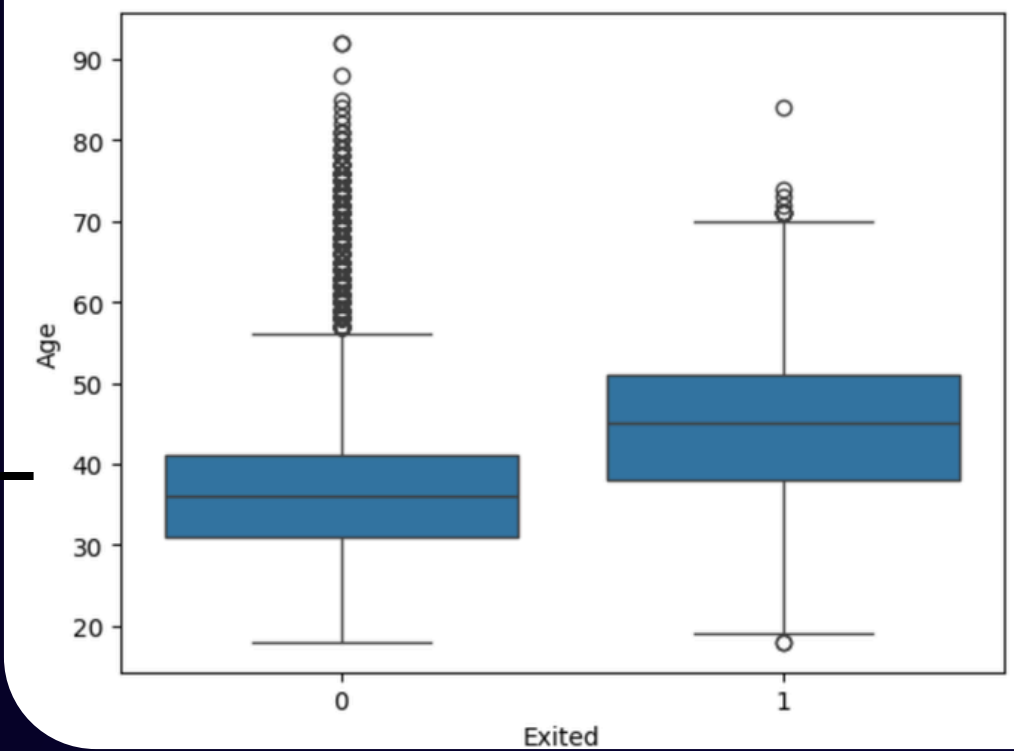
```
sns.countplot(x='IsActiveMember', hue='Exited', data=df)
```

<Axes: xlabel='IsActiveMember', ylabel='count'>

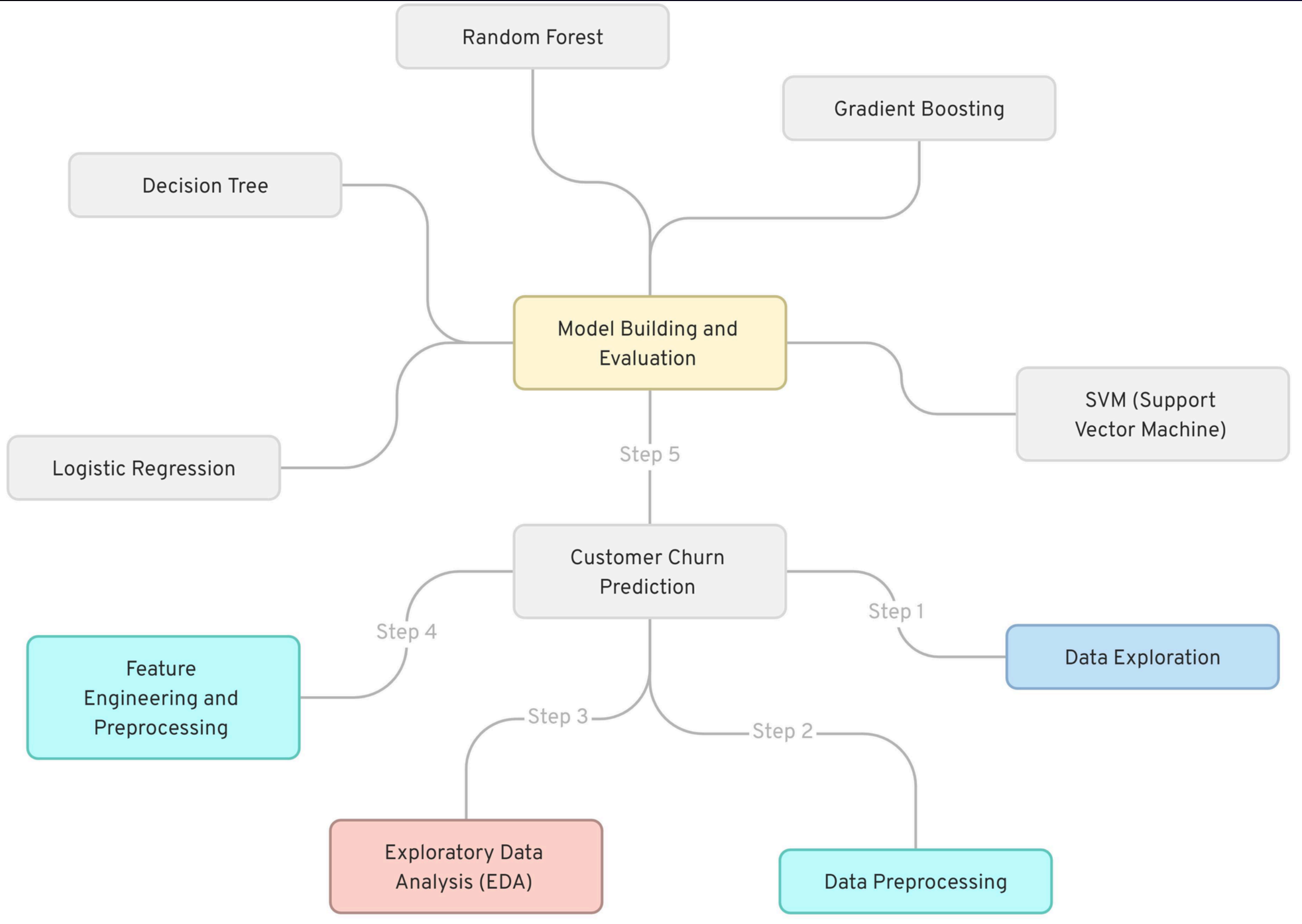


```
sns.boxplot(x='Exited', y='Age', data=df)
```

<Axes: xlabel='Exited', ylabel='Age'>



- Older customers churn more
- Inactive users show high churn probability
- High-balance customers are vulnerable.



Raw Data
↓
Preprocessing
↓
Model Training
↓
Prediction Engine
↓
Retention Recommendation

Models Used

- **Logistic Regression**
- **Balanced Logistic Regression**
- **Random Forest**
- **XGBoost**

Accuracy

Overall correctness.

Precision

Correctness of churn predictions.

Recall

Ability to catch actual churners.

F1 Score

Balance between precision & recall.

ROC-AUC

Ability to separate churners from loyal customers.

Comparison :

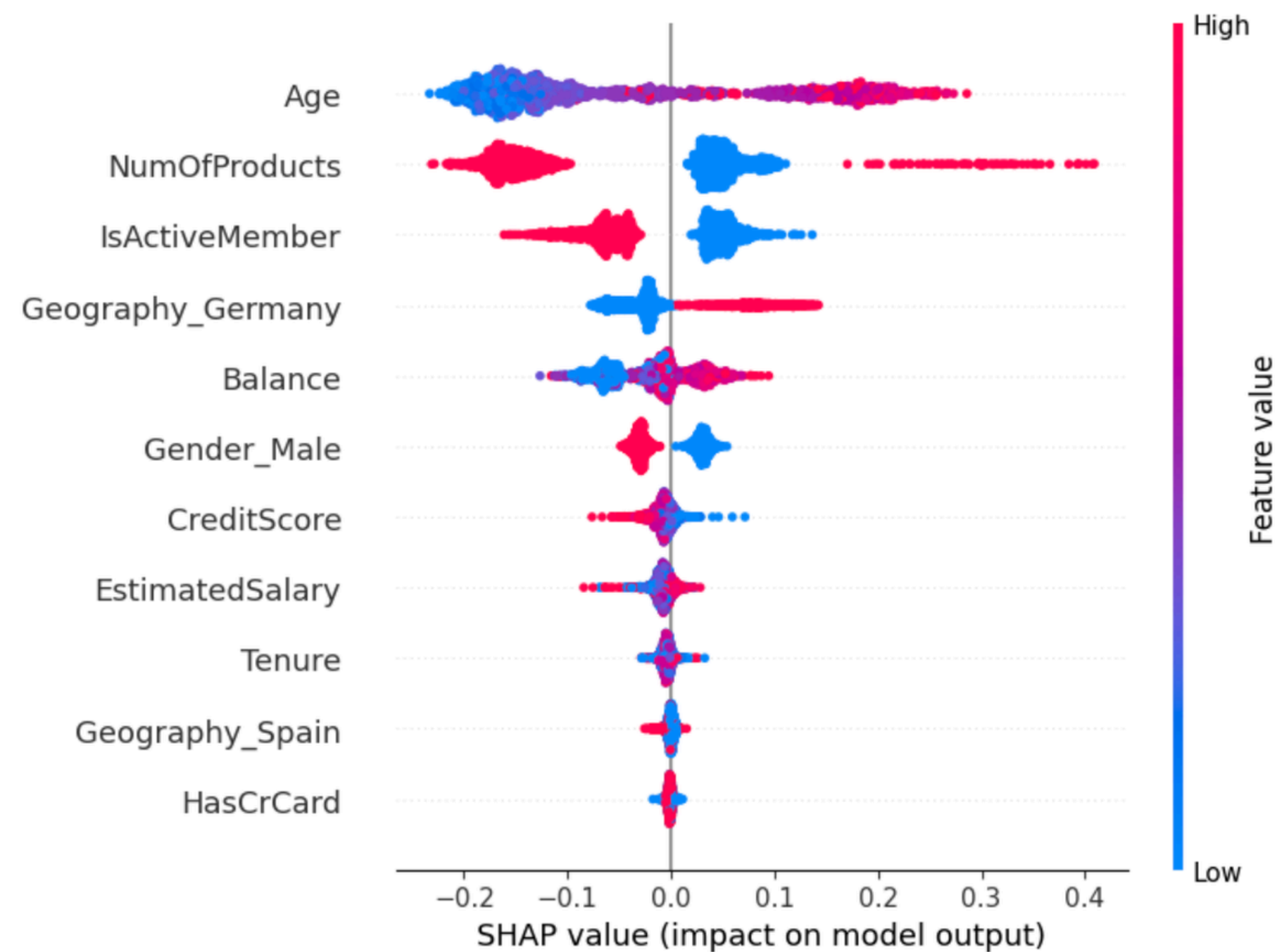
Model	Accuracy	Recall	F1	ROC-AUC
LR	80.8	0.186	0.283	0.774
Balanced LR	71.35	0.7	0.498	0.777
RF	82.6	0.712	0.625	0.865
XGBoost	82.5	0.705	0.621	0.852

Result :

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Higher recall, better churn detection, smarter retention — Random Forest checked all the boxes

SHAP PLOT

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This plot is used to show dependency of the output on the input columns . It gives a sense of both positive and negative dependency of the columns on the model.

- Older customers showed a higher tendency to churn, possibly due to changing financial needs, dissatisfaction with traditional banking experiences, or migration towards competitors offering better premium services.
- Customers with multiple banking products were also more likely to churn, which may indicate product complexity, hidden charges, poor cross-service experience, or reduced satisfaction despite deeper engagement with the bank.
- Inactive members emerged as one of the strongest churn indicators, suggesting that low engagement, infrequent transactions, and weak customer-bank interaction significantly increase the risk of customer loss.
- Customers with higher account balances were found to be more vulnerable to churn as well, which is especially critical because losing such customers directly impacts deposits, investments, and overall bank revenue.

- *Older customers showed higher churn probability, indicating that many long-term or senior customers may feel disconnected from rapidly digitized banking systems. Instead of generic offers, banks should focus on assisted digital onboarding, dedicated relationship managers, simplified interfaces, and trust-driven premium support programs tailored for older demographics.*
- *Customers with a higher number of banking products also displayed elevated churn risk, suggesting that cross-selling alone does not ensure retention. This may point toward product fatigue, fragmented user experience, or dissatisfaction across multiple services. Banks should therefore focus on integrated service ecosystems, unified customer support, and personalized product optimization rather than aggressively increasing product count.*

- *Inactive customers emerged as one of the strongest churn indicators, implying that declining engagement acts as an early behavioral signal before actual churn occurs. Banks can leverage this by implementing activity-triggered interventions such as personalized nudges, transaction-based rewards, dormant account alerts, and predictive engagement campaigns before the customer fully disengages.*
- *High-balance customers were also found to be more likely to churn, which is especially concerning because these customers contribute significantly to deposits and revenue. This suggests that financially valuable customers may have higher expectations regarding service quality, investment support, and personalization. To retain them, banks should prioritize white-glove servicing, exclusive financial advisory programs, premium loyalty benefits, and proactive issue resolution strategies.*

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Thank You